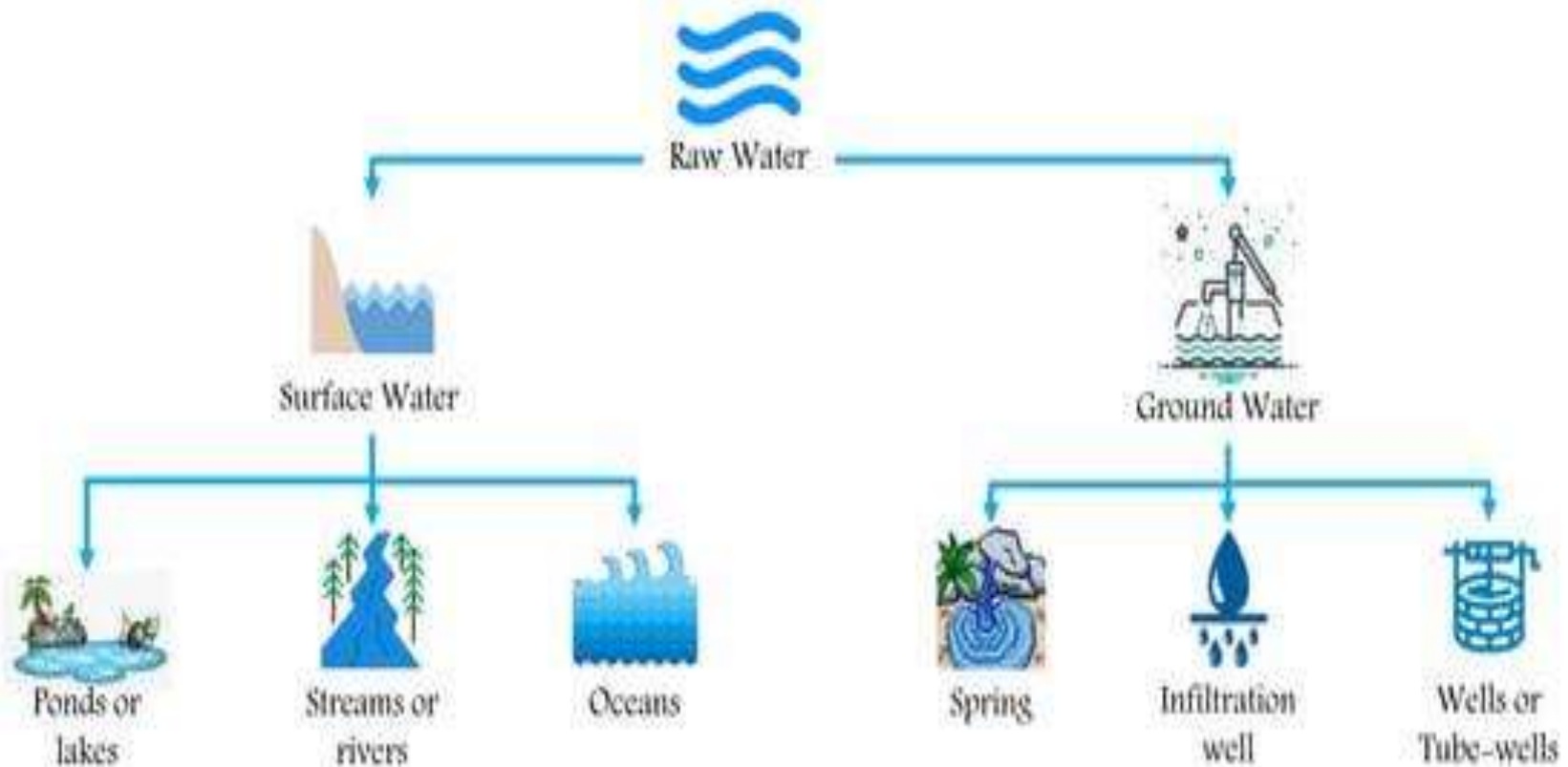


Water Quality & Sources

~SHRUTI BANSAL

Raw Water: Sources



Water Quality

☐ Water Quality is analyzed by,

Physical

- Tells about the appearance
- Related to the Quality of Water
- Analysis is based on physical appearance of water

Chemical

- Represents how water reacts
- Associated with the observed chemical reaction
- E.g. hard and soft water

Biological

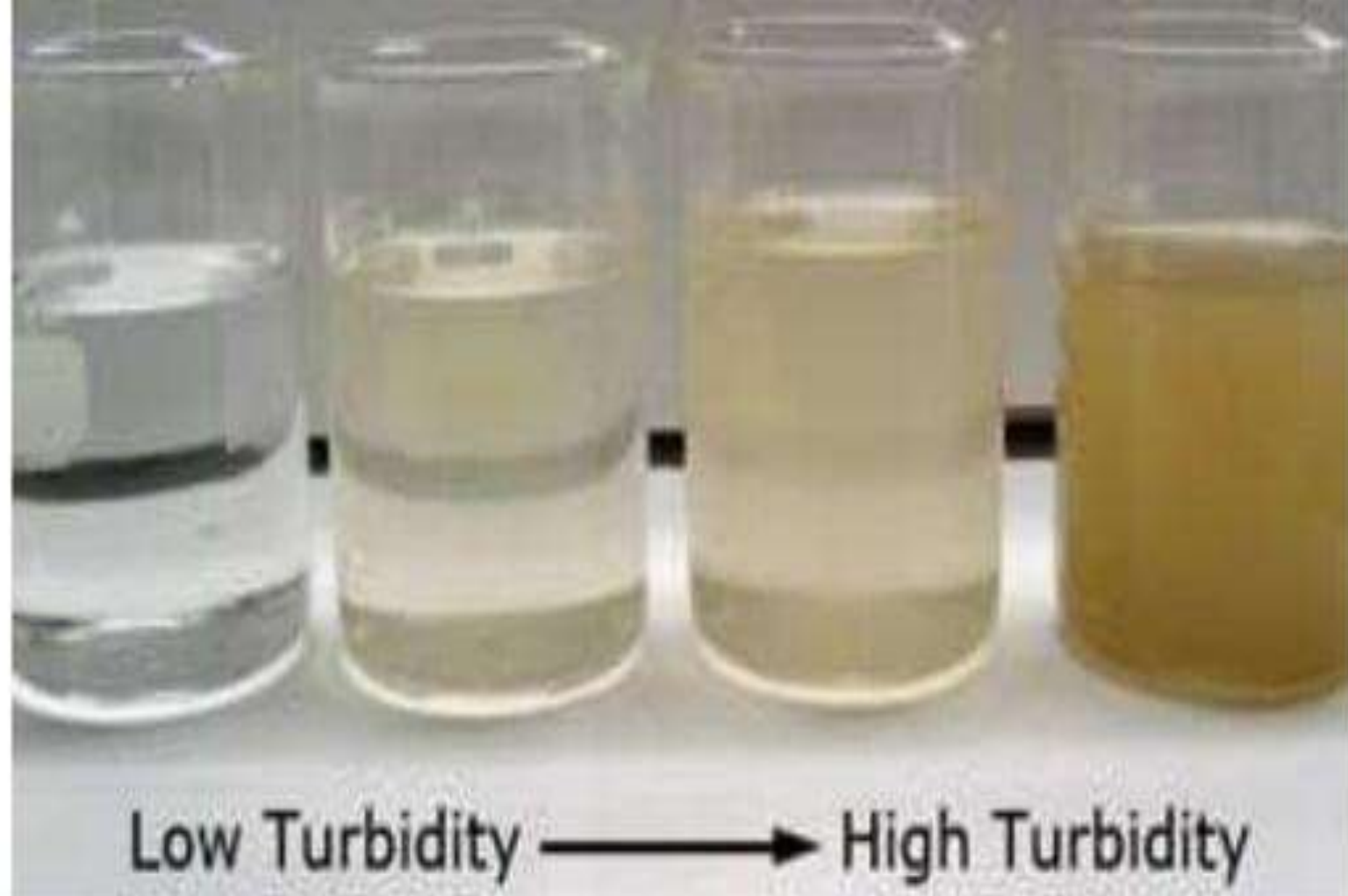
- Signifies the presence of pathogenic bacteria
- Indicates the degree of pollution
- Presence of pathogenic bacteria is signification in terms of public health

Physical Characteristics of Water

- ☐ Turbidity
- ☐ Colour
- ☐ Taste and Odour
- ☐ Temperature

Turbidity

- ❑ Measure of relative clarity of liquid
- ❑ Represents the amount of suspended particles in water
- ❑ Turbidity is measured using Nephelometry
- ❑ It is an optical characteristic of water
- ❑ It is a measurement of the amount of light that is scattered by material in the water when a light is shined through the water sample.
- ❑ The higher the intensity of scattered light, the higher the turbidity.
- ❑ The unit of expression is Nephelometric Turbidity Unit (NTU).
- ❑ The IS values for drinking water is 10 to 25 NTU.



Turbidity of Water

What causes turbidity in water?

- ❑ Material that causes water to be turbid include
 - clay,
 - silt,
 - very fine inorganic and organic matter,
 - algae,
 - dissolved coloured organic compounds, and
 - plankton and other microscopic organisms.

Pure water has zero turbidity

Why turbidity is bad for water?

- ❑ High turbidity deteriorates the aesthetic quality of lakes and streams, having a harmful impact on recreation and tourism
- ❑ Turbidity can also inhibit photosynthesis by blocking sunlight. This impacts the survival of photosynthetic plants and also decreases the Dissolved Oxygen
- ❑ Increase in turbidity indicates potential pollution. The source could be dissolved metals, or pathogens. Either way water consumption with high turbidity can impact human health

Color

- ❑ Primarily a concern because of aesthetic reasons
- ❑ Sources could be:
 - dissolved organic matter from decaying vegetation or
 - some inorganic materials
- ❑ Colour can indicate the presence of organic substances, such as algae or humic compounds
- ❑ The standard unit of colour is that which is produced by one milligram of platinum cobalt dissolved in one litre of distilled water.
- ❑ The IS value for treated water is 5 to 25 cobalt units.

Pure water is colorless

Why colored water is not acceptable?

- ❑ Coloured water is not aesthetically acceptable to the general public.
- ❑ Highly coloured water is unsuitable for
 - laundering.
 - dyeing.
 - paper making.
 - beverage manufacturing.
 - dairy production and
 - other food processing and textile and plastic production.
- ❑ Thus, the colour of the water affects its marketability for both domestic and industrial use

Taste and Odour

- ❑ Water gets taste and odour from many substances in nature or after human use. These substances include:
 - minerals, metals and salts from the soil, and
 - products from biological reactions and constituents of waste water.
- ❑ Water tastes bitter when contaminated with alkaline impurities and salty when the impurities are metallic salts.
- ❑ Biological decomposition of organic debris impart a characteristic taste and odour of rotten eggs which is mainly due to hydrogen sulphide.
- ❑ Growth of algae, micro-organisms, hydrogen sulphide and ammonia give an obnoxious odour to water making it unfit for use.

Pure water is tasteless and odorless

What impact water with odor and taste has?

- ❑ Water with a taste and odour is not aesthetically acceptable to the general public
- ❑ Odours posed by organic substances may pose more than a problem of simple aesthetics since some of those substances may be carcinogenic.

Temperature

- ❑ The ideal temperature of water for drinking purposes is *5 to 12 °C*
- ❑ Temperature has an effect on most chemical reactions that occur in natural water systems.
- ❑ Increase in temperature decreases solubility of Oxygen
- ❑ Increase in temperature increases the growth rate of aquatic microorganism, so they consume dissolved O₂ faster and level of dissolved O₂ decreases.
- ❑ Similarly, temperature affects disinfection process because efficiency of disinfection is lower at lower temperature.

How temperature impacts water quality

- ❑ The toxicity of chemical pollutants increases with increased in temperature.
- ❑ Growth of algae is accelerated and becomes problematic when the algal cells cluster to form mats.
- ❑ Affects disinfection process